

Problem 10.16

Fund A earns interest at a nominal rate of interest of 12% compounded quarterly. Fund B earns interest at a force of interest δ . John invested \$1,000 in each fund five years ago. Today, the amount in Fund A is equal to the amount in Fund B . Calculate δ .

Solution

For Fund A , the quarterly interest rate is

$$\frac{i^{(4)}}{4} = \frac{0.12}{4} = 0.03.$$

There are $4(5) = 20$ quarterly compounding periods in five years. Therefore, the accumulated value of Fund A is

$$1000(1.03)^{20}.$$

Fund B accumulates continuously at force of interest δ . Its value after five years is

$$1000e^{5\delta}.$$

Since the two funds have equal values today,

$$1000(1.03)^{20} = 1000e^{5\delta}.$$

Canceling 1000 and taking natural logarithms gives

$$\begin{aligned}(1.03)^{20} &= e^{5\delta}, \\ 20 \ln(1.03) &= 5\delta.\end{aligned}$$

Thus,

$$\delta = \frac{20 \ln(1.03)}{5} = 4 \ln(1.03) \approx 0.118235204.$$

Therefore, the force of interest is

$$\boxed{\delta \approx 0.118235204}$$

per annum, or approximately

$$\boxed{11.8235\%}.$$